

Angles of a Polygon Using Diagonals

Geometry Worksheet · Grade 7–10

Name: _____

Date: _____

Learning Objectives

- Calculate the number of diagonals in a polygon using the diagonal formula
- Determine the sum of interior angles of a polygon using triangles formed by diagonals
- Find individual interior and exterior angle measures of regular polygons

Problems

1. How many diagonals can be drawn from a single vertex of a regular quadrilateral (square)?

2. How many triangles are formed when all diagonals are drawn from one vertex of a regular quadrilateral (4 sides)?

3. What is the sum of interior angles of a regular quadrilateral using the triangles formed by its diagonals?

$$S = (n - 2) \times 180^\circ$$

4. Find the measure of each interior angle of a regular quadrilateral (square).

$$\text{Each angle} = \frac{S}{n}$$

5. Using the diagonal method, find the sum of interior angles of a regular pentagon (5 sides).

$$S = (n - 2) \times 180^\circ$$

6. Find each interior angle of a regular pentagon using the diagonals method.

$$\text{Each angle} = \frac{(n - 2) \times 180^\circ}{n}$$



$$D = \frac{n(n-3)}{2}$$

8. A regular polygon has an interior angle sum of 1080 degrees. How many sides does it have, and what is each interior angle?

$$n = \frac{S}{180^\circ} + 2$$

9. A regular polygon has 20 diagonals. Determine the number of sides of the polygon, then find the sum of its interior angles and the measure of each interior angle.

$$D = \frac{n(n-3)}{2}$$

10. A regular polygon has 35 diagonals. Find the number of sides, the sum of interior angles, each interior angle, and each exterior angle.

$$D = \frac{n(n-3)}{2}, \quad \text{Exterior angle} = \frac{360^\circ}{n}$$



Angles of a Polygon Using Diagonals — Answer Key

Geometry Worksheet · Grade 7–10

Answer Key

1. Answer: 1 diagonal

- A diagonal connects two non-adjacent vertices.
- A square has 4 vertices. From one vertex, you can connect to $4 - 1 - 2 = 1$ non-adjacent vertex.
- So exactly 1 diagonal can be drawn from a single vertex.

2. Answer: 2 triangles

- When diagonals are drawn from one vertex of an n -sided polygon, the number of triangles formed is $n - 2$.
- For a quadrilateral: $4 - 2 = 2$ triangles.

3. Answer: 360°

- Drawing a diagonal from one vertex divides the quadrilateral into 2 triangles.
- Each triangle has an angle sum of 180° .
- Total interior angle sum $= 2 \times 180^\circ = 360^\circ$.

4. Answer: 90°

- The sum of interior angles of a quadrilateral is 360° .
- A regular quadrilateral has 4 equal angles.
- Each angle $= 360^\circ \div 4 = 90^\circ$.

5. Answer: 540°

- Draw diagonals from one vertex of the pentagon — this creates $5 - 2 = 3$ triangles.
- Sum of interior angles $= 3 \times 180^\circ = 540^\circ$.

6. Answer: 108°

- Sum of interior angles of a pentagon $= (5 - 2) \times 180^\circ = 540^\circ$.
- A regular pentagon has 5 equal angles.
- Each interior angle $= 540^\circ \div 5 = 108^\circ$.

7. Answer: 9 diagonals

- Use the formula $D = n(n - 3) / 2$, where $n = 6$.
- $D = 6(6 - 3) / 2 = 6 \times 3 / 2 = 18 / 2 = 9$.
- A regular hexagon has 9 diagonals.

8. Answer: 8 sides; each interior angle = 135°

- Use $S = (n - 2) \times 180^\circ$, so $n - 2 = S / 180^\circ = 1080^\circ / 180^\circ = 6$.
- Therefore $n = 6 + 2 = 8$ sides (regular octagon).
- Each interior angle $= 1080^\circ \div 8 = 135^\circ$.



**9. Answer: 8 sides; interior angle sum = 1080°; each angle = 135°**

- Set $D = n(n - 3)/2 = 20$, so $n(n - 3) = 40$.
- Try $n = 8$: $8 \times (8 - 3) = 8 \times 5 = 40$. ✓ So the polygon has 8 sides.
- Interior angle sum = $(8 - 2) \times 180^\circ = 6 \times 180^\circ = 1080^\circ$.
- Each interior angle = $1080^\circ \div 8 = 135^\circ$.

10. Answer: 10 sides; interior sum = 1440°; each interior angle = 144°; each exterior angle = 36°

- Set $D = n(n - 3)/2 = 35$, so $n(n - 3) = 70$.
- Try $n = 10$: $10 \times (10 - 3) = 10 \times 7 = 70$. ✓ So the polygon has 10 sides (decagon).
- Interior angle sum = $(10 - 2) \times 180^\circ = 8 \times 180^\circ = 1440^\circ$.
- Each interior angle = $1440^\circ \div 10 = 144^\circ$.
- Each exterior angle = $360^\circ \div 10 = 36^\circ$. Check: $144^\circ + 36^\circ = 180^\circ$. ✓

